

Comp 322
Mobile Application Development
Lab 3
Android Studio and Jetpack Compose

The purpose of this lab is to introduce students to creating multiplatform mobile applications using Android Studio and Jetpack Compose. The completed application will display a single random Bible verse (book, chapter, verse and text) to the user as formatted text each time the application is launched. This lab will use the BibleVerse and VerseOfTheDay classes that we created in Lab 2 (see Lab 2 requirements).

- 1) Create a new Android Jetpack Compose (empty activity) project. Name the project lab3. The Android Studio Project should use the following package name convention - com.<studentname>.lab3. Replace <studentname> with your last name.
- 2) Isolate the BibleVerse and VerseOfTheDay classes created in Lab 2 and save them as separate files – BibleVerse.kt and VerseOfTheDay.kt respectively.
- 3) Import them into your Android Studio project. Make sure to place the package name at the top of the file.
- 4) Create a non-composable method in the MainActivity.kt file that will create an instance of the VerseOfTheDay class and retrieve a random Bible verse using a public method from the VerseOfTheDay class.
- 5) Create a composable method that will retrieve the random Bible verse from the method created in step 4.
- 6) Create a preview composable to preview the composable created in step 5. Test the preview composable to verify that it works (include a screen image in the lab report).
- 7) Modify the setContent callback in the MainActivity class to call the composable created in step 5 when the application is run.
- 8) Run the application in an emulator and verify that it is displaying the output correctly. Fix any errors in the program and adjust the text size and line height attributes to display appropriately.

The lab report should contain a brief description of what the lab does followed by screen images of the source code and successful execution. The lab report should also contain an explanation of any problems encountered while completing the lab and the estimated time for completion. Submit the lab report along with the source code to Canvas by the due date. If you have any questions, please ask.